Replication of Moser, Voena, and Waldinger (2014)

You will replicate Sections I and II.A-E and conduct additional analyses beyond those in the paper using the dataset **patent_classes.dta**. Carefully read the paper, review the data description document, and answer the following questions.

1. Research question:
    a. What is the research question of the paper, and why is it important or interesting?
    b. Discuss any potential channels through which the effect of interest could be positive.
    c. Discuss any potential channels through which the effect of interest could be negative.
2. Explain the context of the historical event (natural experiment) that the paper exploits to answer the research question.
3. What is the structure of the dataset? Is it cross-sectional, time-series, or panel data? What are the entities and time periods?
4. Summary statistics:
    a. Replicate Table 1 and Figures 2A and 2B.
    b. Provide the difference-in-differences estimate of the treatment effect of interest based on the summary statistics in Columns (2) and (3), before conducting the formal regression analysis.
5. Ordinary least squares:
    a. Regress the dependent variable of interest on the independent variable of interest, either alone or with control variables, but without fixed effects. The regression equations are the same as Equations (1) and (2), except without the fixed effects terms, as shown below:

$$Patent\ by\ US\ inventors_{c,t} = \alpha_0 + \beta\ emigre\ class_c \cdot post_t + \boldsymbol{\gamma}'\boldsymbol{X}_{c,t} + \epsilon_{c,t}$$
$$Patent\ by\ US\ inventors_{c,t} = \alpha_0 + \beta\ number\ emigre\ patents_c \cdot post_t + \boldsymbol{\gamma}'\boldsymbol{X}_{c,t} + \epsilon_{c,t}$$

        i. Present the results in a table formatted similarly to Table 2, but with "No"s in the "Year fixed effects" and "Class fixed effects" rows.
        ii. Interpret the estimated β's.
        iii. What do the control variables measure, and what do they control for?
        iv. Discuss any remaining potential biases and how they could be addressed.
6. Difference-in-differences:
    a. Now we exploit the fact that the dataset is panel. Obtain the difference-in-differences estimates of the treatment effect of interest using Equations (1) and (2).
        i. Replicate Table 2.
        ii. Interpret the estimated β's.
        iii. Mark the fixed effects terms in the regression equations. What do they control for?
        iv. Discuss any remaining potential biases and how they could be addressed.
    b. Next we will run a specification described in the last paragraph of Section II.A.
        i. The regression equations for this specification are not provided in the paper. Write the regression equations that correspond to Table A1 Columns (1) and (2). What is the omitted category of patent classes in these regressions, and why is it omitted?
        ii. Replicate Table A1 Columns (1) and (2).
        iii. You should have three β's, one for each of "1 émigré patent*post", "2 émigré patent*post", and "3 or more émigré patent*post". Interpret the estimated β's. The interpretation should be relative to the omitted category of patent classes.
    c. Run a version of the difference-in-differences regression given in Equations (3).
        i. Figure 3 visualizes the estimation results. Replicate Figure 3.
        ii. Interpret the estimated β_t's.

d. Now we will allow time trends to differ for each patent class.
    i. Add entity-specific time trends $\eta_c \cdot t$ to Part a, run the regressions, present the results in a table formatted similarly to Table 2, and interpret the estimated β's. Your table should resemble Table 2, but below the "Class fixed effects" row, insert a row labeled "Class-specific time trends" with entries of "Yes".
    ii. Add entity-specific time trends $\eta_c \cdot t$ to Part b, run the regressions, present the results in a table formatted similarly to Table A1, and interpret the estimated β's. Your table should resemble Table A1, but below the "Class fixed effects" row, insert a row labeled "Class-specific time trends" with entries of "Yes".
    iii. Add entity-specific time trends $\eta_c \cdot t$ to Part c, run a regression *similar* to Equation (4), present the results in a figure *similar* to Figure A1, and interpret the estimated β_t's. I said 'similar', not 'same', because your estimation will include years before 1933, as in Equation (3), unlike Equation (4). Accordingly, your figure will include years before 1933, as in Figure 3, unlike Figure A1.

7. Instrumental variables:
    a. What instrument do the authors use? What issues is the instrument addressing? We learned that a valid instrument must satisfy two conditions. Explain why the instrument in the paper is valid.
    b. As was emphasized in the lecture, "If you can't see it in the reduced form, it ain't there."
        i. Replicate Figure 4.
        ii. Provide the difference-in-differences estimate of the reduced-form effect based on the summary statistics in Table 1 Columns (4) and (5), before conducting the formal regression analysis.
    c. Run the first-stage and reduced-form regressions using Equations (5) and (6) (and analogous equations using the number of pre-1933 patents by dismissed chemists in each patent class), replicate Table 3, and interpret the estimated ϕ's and β's.
    d. Table 3 reports the first-stage F-statistics. What do they test, and what are the test results?
    e. Run the instrumental variables regression using the suggested strategy.
        i. Replicate Table 4.
        ii. Interpret the estimated β's.
        iii. Discuss any remaining potential biases and how they could be addressed.
    f. Now we will allow time trends to differ for each patent class.
        i. Add entity-specific time trends $\eta_c \cdot t$ to Part c, present the results in a table formatted similarly to Table 2, and interpret the estimated β's. Your table should resemble Table 3, but below the "Class fixed effects" row, insert a row labeled "Class-specific time trends" with entries of "Yes".
        ii. Add entity-specific time trends $\eta_c \cdot t$ to Part e, present the results in a table formatted similarly to Table 2, and interpret the estimated β's. Your table should resemble Table 4, but below the "Class fixed effects" row, insert a row labeled "Class-specific time trends" with entries of "Yes".
8. Assume the authors stopped here. If you were a referee, what would you suggest the authors do more or check further?
9. Summarize Section III of the paper. What additional questions are the authors addressing, and how do they address them? (Explain the data and methodologies.) What are the results?

Instructions:

1. The paper, appendix, and dataset for this assignment are available on Brightspace.
2. On the publication webpage (https://doi.org/10.1257/aer.104.10.3222), you can find the replication package provided by the authors. You may refer to it, but do not simply reuse the code without careful review, as some parts are inefficient, incomplete, or contain minor errors.
3. For questions that require you to generate results (e.g., tables, figures, or numerical estimates), include the code you used directly below your answer and explain each line of the code (even if it feels repetitive). You may use any statistical software.
4. Submit two files: (i) a document that contains your answers to the questions above (including the code and explanations as described in Instruction 3) and (ii) an executable code file (e.g., .do or .py) that generates the results reported in your answers.
5. When it says "replicate a table", your table should present the same content as in the paper, but it does not need to match the exact formatting. You are not required to generate the table directly using statistical software; you may create it, for example, in Word. Do not include screenshots of tables from the paper.
6. When it says "replicate a figure", you should generate the figure using your preferred software. Your figure should present the same content as in the paper, but be slightly different in formatting from the original (e.g., by using different colors) to demonstrate that you have created it yourself. Do not copy or screenshot figures from the paper.